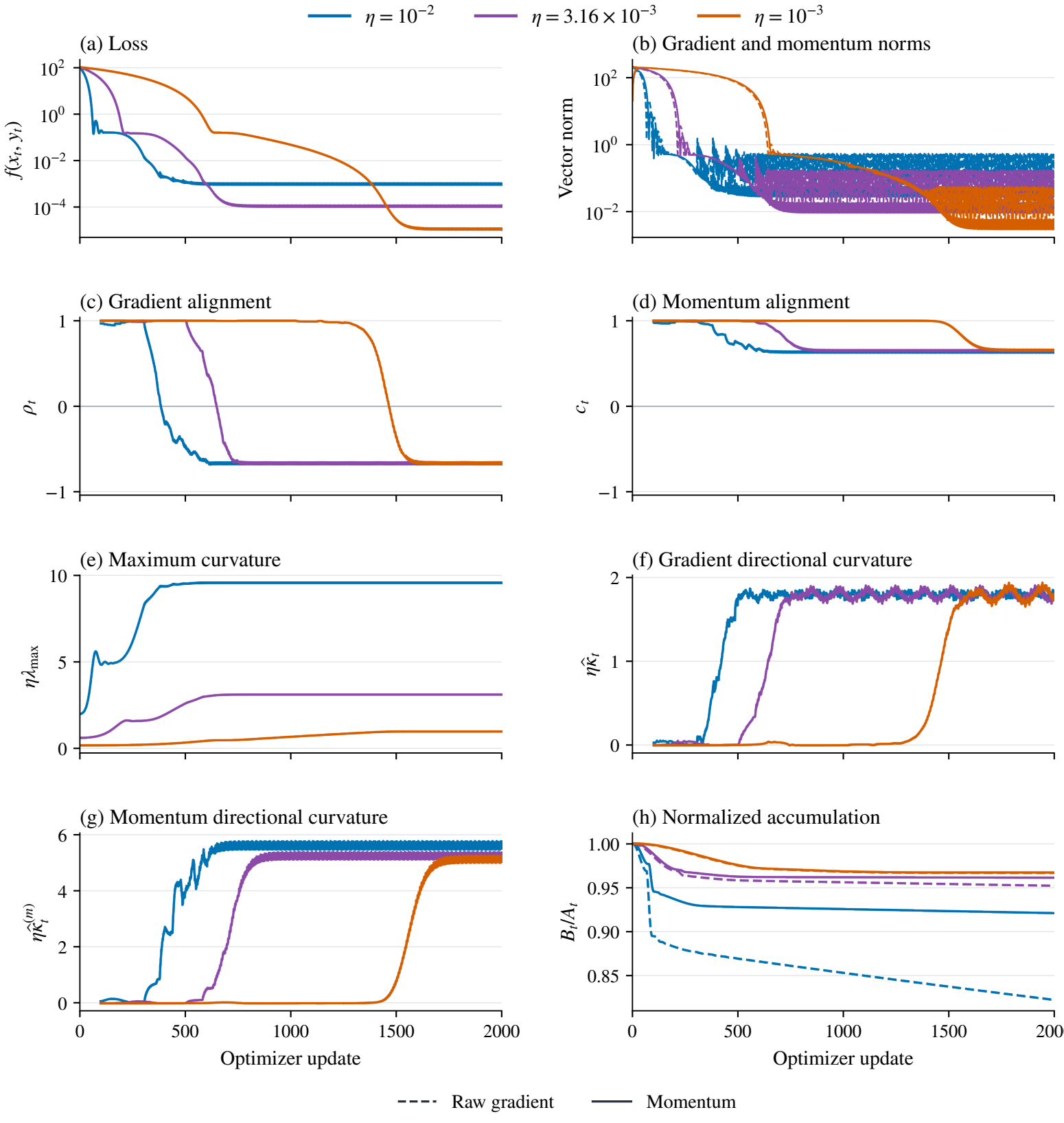

$\eta = 10^{-2}$
$\eta = 3.16 \times 10^{-3}$
$\eta = 10^{-3}$
(a) Loss
$f(x_t, y_t)$
(b) Gradient and momentum norms
Vector norm
(c) Gradient alignment
ρ_t
(d) Momentum alignment
c_t
(e) Maximum curvature
$\eta\lambda_{\max}$
(f) Gradient directional curvature
$\eta\hat{\kappa}_t$
(g) Momentum directional curvature
$\eta\hat{\kappa}_t^{(m)}$
(h) Normalized accumulation
B_t/A_t
Optimizer update
Optimizer update
Raw gradient
Momentum